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# Boosting Practical (Cross Validation + AdaBoost)
from sklearn.datasets import load_iris
from sklearn.model_selection import cross_val_score
from sklearn.ensemble import AdaBoostClassifier
from sklearn.tree import DecisionTreeClassifier
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# Load Dataset
data = load_iris()
X = data.data
y = data.target
```

```
# (a) CROSS VALIDATION
print("---- Cross Validation ----")

model = DecisionTreeClassifier(max_depth=1)

cv_scores = cross_val_score(model, X, y, cv=5)

print("Cross Validation Scores:", cv_scores)
print("Average Accuracy:", cv_scores.mean())
```

```
---- Cross Validation ----
Cross Validation Scores: [0.66666667 0.66666667 0.66666667 0.66666667 0.66666667]
Average Accuracy: 0.6666666666666666
```

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# (b) ADABOOST IMPLEMENTATION
print("\n---- AdaBoost ----")

ada_model = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(max_depth=1),
    n_estimators=50,
    learning_rate=1
)

ada_model.fit(X, y)

accuracy = ada_model.score(X, y)

print("AdaBoost Accuracy:", accuracy)
```

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---- AdaBoost ----
AdaBoost Accuracy: 0.98
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